

A Crypto Observation Layer for Public LLMs: Compiling World Bundles with Quality-Gated Refusal

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Abstract

Claim. Public LLMs fail on crypto markets for want of a reliable *observation layer*, not for want of another trading strategy. We contribute that layer as an anonymized *world-model runtime* (state compiler + quality gate, not a generative simulator): asynchronous multi-band evidence is compiled into a point-in-time (PIT) *world bundle* that public models can consume; when support is thin, a hard refuse flag forces abstention so half-baked context cannot drive decisions. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{AI}$; completeness, honesty (B, U, H) , and WMI/ACWMI make quality machine-readable; evidence IDs bind actions to disclosed fields. **Evidence.** On identical OOS asset-days, a Compiled bundle makes the market legible: grounding recovers ready/missing/tilt at ≈ 0.97 vs. $\lesssim 0.23$ under Raw. The same layer’s quality gate is enforceable: across four live public LLMs (temperature 0), Compiled abstains on every thin day (1.0); Ungated (same content, soft text) abstains only ≈ 0.68 ; Raw over-trades and loses; Blind refuses with no feed. Full-OOS and band-thick splits reproduce the ranking. This panel never exceeds the production refuse threshold (max WMI $\approx 0.093 < 0.2$), so Compiled’s full refusal is a scarce-support stress test of the gate, not a claim the layer never opens. On the open slice (WMI ≥ 0.05 ; $n=1224$), live Ungated CE is positive for every vendor (four-vendor mean $\approx +0.29$; block-bootstrap mean $\approx +0.31$, CIs include 0) and a no-LLM rule beats momentum ($\Delta CE=0.315$, $p=0.034$)—a content lower bound for the compiled fields, not Compiled-open trading (the frozen prompt can still force abstain via a separate thinness clause). **Falsifiable claim.** Compilation makes disclosed world fields verifiable; hard quality gates enforce thin abstention where soft text does not; open-slice fields are economically nonempty under a no-LLM probe.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

observation layer, world bundle, quality-gated refusal, public LLMs, cryptocurrency, point-in-time, trustworthy AI, world models, systems

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1 Introduction

Core claim (read this first). Public LLMs fail on crypto *before* strategy: they lack a complete, honest, auditable *market observation layer*. Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [13, 15], while generative world models [7, 8, 11] and tool-using finance agents [22–24] typically *assume* a usable observation stream. This paper builds that missing layer—not a new trading agent and not a generative simulator. In one line: *give public LLMs a crypto observation layer that compiles asynchronous evidence into a consumable PIT world bundle, and forces abstention when quality is insufficient so half-baked context cannot harm decisions.*

What we build. We implement the layer as an anonymized *market world-model runtime*: a state compiler from raw multi-band evidence to a PIT-safe *world bundle* for GPT /DeepSeek /GLM /Gemini-class models, with a typed quality gate (should_ai_abstain) as the refuse interface. “World model” here means *observation compiler + quality-gated runtime* (Section 3), not a next-state dynamics model. The compiled world must be **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Refusal is the safety valve of that layer, not a substitute for seeing the market.

How we evaluate. The layer has two faces on identical OOS days; experiments report the gating ranking first (RQ1), then seeing (RQ2), then content (RQ3). **RQ1 (gating; primary ranking):** When the world is thin, does a hard quality flag *enforce* abstention across live public LLMs, where soft disclosure does not? Four arms—Compiled (hard flag), Ungated (same content, no flag), Raw (momentum fragment), Blind (no feed). **RQ2 (seeing):** Does the compiled bundle make analyst answers (ready /missing /tilt) verifiable against PIT ground truth? **RQ3 (secondary):** Are the vintaged tilts economically nonempty under a transparent no-LLM rule vs. momentum /buy-and-hold?

Contributions. (1) **Semantics:** epistemic observations, Π_t , lag bounds, completeness/honesty, WMI gates, evidence-bound actions (interface semantics, not a pricing theorem). (2) **Runtime:** vintage collectors, BandPIT previous-close clock, quality-tagged world bundles, shocks O_t , multi-vendor adapter, frozen four-arm protocol. (3) **Live validation:** grounding ≈ 0.97 vs. Raw 0.04–0.23; hard-gate thin-abstain 1.0 vs. Ungated ≈ 0.68 ; Raw over-trades; Blind refuses; open-slice Ungated CE is a content lower bound, not Compiled-open trading. Estimand: four-arm ranking, not ΔCE vs. a strongest agent.

2 Related Work

World models in AI. World models learn compact states and often dynamics [7, 8, 11]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics

or LLM policy. We claim the missing observation layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [14], open financial LLMs [21], memory-augmented trading agents [12, 23], and tool-augmented agents [22, 24] improve agent skill and orchestration. Recent ICAIF work targets adjacent layers: FinAgentBench evaluates agentic retrieval for financial QA [1]; FinSearch builds temporal-aware search agents over market and news feeds [18]; FinResearchBench judges long-horizon research agents with logic-tree protocols [19]; FactorMAD uses multi-agent debate for interpretable alpha mining [3]. Those systems improve how agents fetch, debate, and are scored, but still assume a usable observation stream. We study the missing prior step—a PIT-safe *observation layer* that compiles a world bundle and quality-gates action—and evaluate within-model Compiled / Ungated / Raw / Blind under a frozen schema rather than proposing another memory router, retrieval planner, or debate society.

Selective prediction and trustworthy AI. Abstention can be optimal under noise [4, 5]. We treat refusal as the quality gate of an observation layer, tied to world quality via an explicit runtime field, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [13, 15] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [9, 17, 20]. Asset-pricing ML [6, 10, 16] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Observation-Layer Semantics

This section defines the observation layer formally (interface semantics, not a pricing theorem): what counts as an observation, how compilation yields a world state, and when quality forces refusal. Readers who want the systems object first may skim to Section 4; the empirical estimand is the four-arm protocol in Section 5.

Definition 3.1 (Epistemic observation). An epistemic observation is

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}) :$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 3.2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Definition 3.3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled layer exposes the quality-gate flag (should_ai_abstain) and thin_world as first-class fields of the world bundle.

Assumption 3.4 (Bounded lag and increments). Band j has max observation lag Δ_j ; latent state increments satisfy $\|S_u - S_{u-1}\| \leq \tilde{\delta}$.

PROPOSITION 3.5 (LAG RECONSTRUCTION BOUND). *Under Assumption 1,*

$$\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\tilde{\delta} + C_{\text{noi}}\epsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.5 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

PROPOSITION 3.6 (COMPILATION $\neq$ FEATURE DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

PROPOSITION 3.7 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

PROPOSITION 3.8 (LOBO CONTENT VS. GATING). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Implication for the runtime. Props. 3.6–3.7 say the observation layer must export gates and abstain flags, not tilts alone: compilation without quality disclosure is still a feature dump; disclosure without a hard gate still lets thin worlds drive action. Prop. 3.8 motivates the economic check: if deleting a durable band does not change actions, the bundle is empty of content; the probe shows that is not the case here.

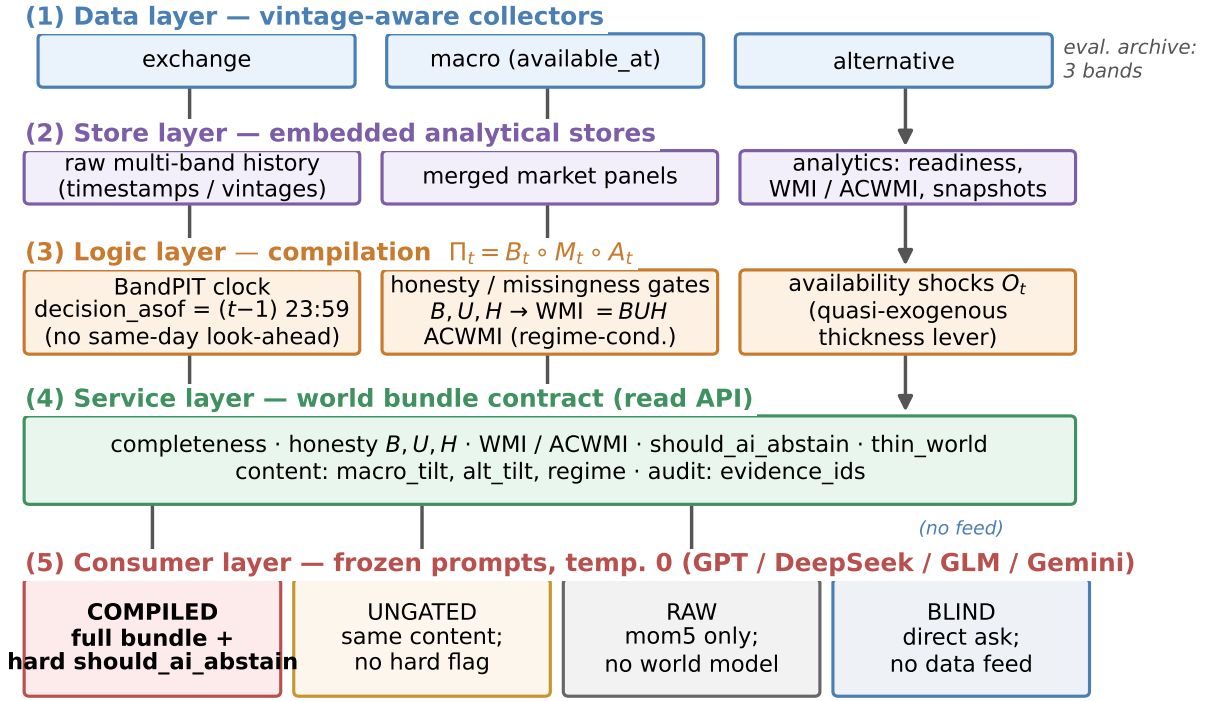
REMARK 3.9 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver an observation layer: state compiler + quality-gated refuse runtime.*

4 Runtime Architecture

We describe the anonymized prototype as a layered system (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

- (1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro available_at vintages.
- (2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI / ACWMI, paper world snapshots).
- (3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty and missingness gates plus band assembly implement Π_t ; world-quality indices and availability shocks O_t are first-class.
- (4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
- (5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, Raw, or Blind treatments.



Live abstention outcomes: Sec. ~Experiments (Table / four-arm figure).

Figure 1: Anonymized observation-layer runtime. Five numbered stages map raw multi-band evidence to a consumable world bundle; the consumer stage defines the frozen four-arm protocol (Compiled / Ungated / Raw / Blind). Live seeing and gating outcomes are in Sec. 5; Blind has no feed from the service stage.

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t , WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: the evaluation archive populates three public bands—exchange klines, funding, and open interest via REST; daily macro risk indices (equity volatility, dollar index) with available_at vintages; and aggregate stablecoin

circulation—each row stored with its observation timestamp. *Processing*: for every asset-day, BandPIT selects the latest observation with timestamp $\leq (t-1) 23:59$, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t . *Outputs*: a ~3,990-row PIT panel (10 assets $\times$ 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

4.3 PIT previous-close clock

For calendar day t , the decision information set is reconstructed at $(t-1) 23:59$. The payoff is the same-calendar-day close-to-close return r_t . This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in decision_asof, and every live run pins timing_protocol to the previous-close protocol.

4.4 World bundle (consumer object)

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, should_ai_abstain=true, and evidence_ids

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetches (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H , WMI, ACWMI, should_ai_abstain, thin_world, O_t , evidence_ids

Table 3: World-bundle field groups (observation layer output).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	n_{ready} /limited/missing, band statuses
Honesty	B, U, H , main-view / stale gates
Quality	WMI, ACWMI, should_ai_abstain, thin_world
Content	macro_tilt, alt_tilt, regime, cascade
Audit	evidence_ids, EAR required

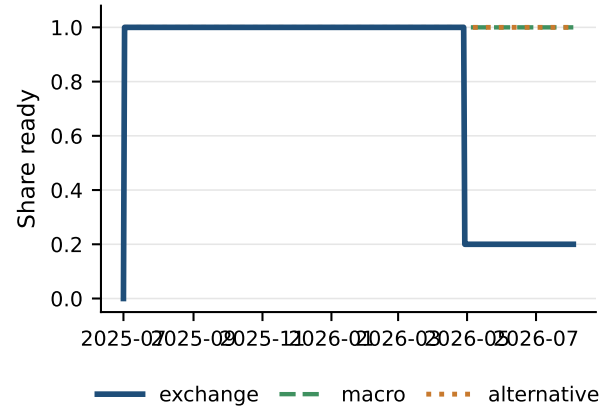
binding actions to disclosed bands/tilts. The bundle is what the LLM sees; the hard boolean is how the layer *gates* action when that view is thin. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only mom5 (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  },
  "world_model_index": {
    "wmi": 0.015,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
  "audit": {
    "evidence_ids": [
      "band:exchange:missing",
      "band:macro:ready"
    ]
  }
}
```

4.5 Four information arms and LLM adapter

- **Compiled (full observation layer):** full world bundle; frozen prompt *requires* abstain when should_ai_abstain is true.
- **Ungated (disclosure without the gate):** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness (seeing without enforcement).
- **Raw (thin integration):** mom5 only; no quality index or abstention guidance. This is a common price-only wiring, not an exhaustive tool-agent baseline: the primary estimand

**Figure 2: PIT readiness for the three-band evaluation archive (solid / dashed / dotted for grayscale).**

is the four-arm ranking of the observation layer (hard gate vs. soft vs. fragment vs. none), not Δ CE against the strongest possible agent.

- **Blind (direct ask):** “Today is d . How should I trade a ?” with *no* data feed.

Actions $\in \{\text{bullish, bearish, neutral, abstain}\}$. Primary metrics are thin-world abstain rate and EAR proxy; CE is reported as avoided loss under refusal, not as an alpha claim. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive *band-thick* contrast below. The refuse rule sets should_ai_abstain when $WMI < 0.2$. On this scarce OOS panel max $WMI \approx 0.093$, so every day is thin and Compiled never opens—a scarce-support stress test of the quality gate, not a claim the observation layer never allows action. The adapter uses OpenAI-compatible chat completions; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt protocols

Listings 2–3 excerpt the frozen Compiled and Ungated prompts (temperature 0; action schema identical). Both consume the same world-bundle content; the only intentional difference is hard vs. soft quality gating. Raw prompts omit world-quality language entirely.

Listing 2: Compiled observation layer (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft guidance (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why the hard quality gate is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the observation layer’s job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.7). Ungated measures soft judgment from the same disclosed bundle; Raw measures the common price-signal-only wiring. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the gate.

4.7 Service surface and reproducibility

The service layer exposes read-only objects for the consumer harness: quality-tagged bundles, readiness and WMI/ACWMI snapshots, shock queries O_t , and health metadata. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and gateway URL are configuration, not estimands. Credentials never enter artifacts. Frozen prompts, protocol notes, and table builders ship with the submission; an anonymized harness is available to reviewers on request and public on acceptance.

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days ($\approx 2,000$ OOS asset-days). Live RQ1 uses the *same* stratified 100 OOS asset-days across arms (seed fixed), temperature 0, frozen prompts, and an OpenAI-compatible gateway; a checkpointed full-OOS sweep tests population scale. Vendor models are GPT-, DeepSeek-, GLM-, and Gemini-class flash/mini IDs listed in Table 4. Scarce/thin labels bind on OOS by design; Sec. 5.4 covers fully populated days (the no-LLM probe uses 199 OOS days after one-day return alignment).

5.2 RQ1 (primary ranking): Quality gate of the observation layer

Table 4 and Figure 3 report the gating face of the layer; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world-bundle fields on 93–99% of days. This is quality-gate obedience by design, not spontaneous thinness discovery. Under Ungated, the *same* disclosed bundle without the hard flag yields mean thin-abs. 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and gemini-3.5-flash-lite falls to 0.43 and trades. Unconditional Ungated CE (abstain days as cash) spans -1.04 to $+0.02$; restricting to days the model actually acts, three of four vendors post large negative act-conditional CE (mean ≈ -1.9), so soft judgment does not safely convert a half-gated observation into profitable action. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Mean Δ CE (Compiled–Raw) is $+1.07$ and positive for every vendor, because the gated layer sits out losses that Raw incurs on sparse support.

Full-OOS replication. To check that the 100-day ranking is not a subsample artifact, we checkpointed the three data arms on the *full* OOS panel ($n=2000$ asset-days; every OOS day is thin-world under the scarce-support design). Compiled thin-abstain is 1.000 [0.998, 1.000] for all four vendors. Ungated rates tighten the Wilson intervals, with mean thin-abs. 0.56 (GPT 0.666 [0.645, 0.686]; GLM 0.679

Table 4: Live abstention rates on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. Δ CE is avoided loss, not an alpha claim.

Model	Thin C	Thin U [CI]	Abs. R [CI]	Δ CE
gpt-5.4-mini	1.00	0.63 [0.53, 0.72]	0.04 [0.02, 0.10]	+0.73
deepseek-v4-flash	1.00	0.81 [0.72, 0.87]	0.75 [0.66, 0.82]	+1.22
glm-5.2	1.00	0.86 [0.78, 0.91]	0.11 [0.06, 0.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [0.34, 0.53]	0.07 [0.03, 0.14]	+1.30
Mean	1.00	0.68	—	+1.07

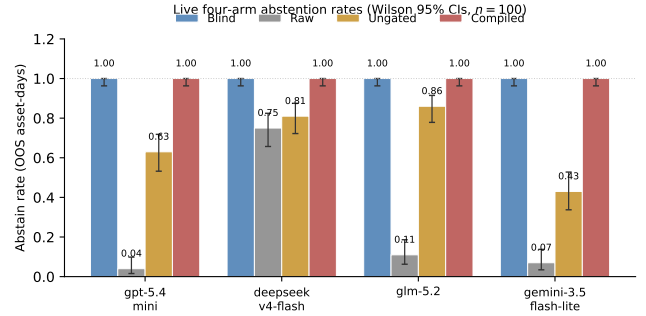


Figure 3: Live four-arm abstention rates (same 100 days; Wilson 95% CIs). Blind refuses with no feed; Raw over-trades on a fragment; Ungated soft-judges a disclosed bundle; Compiled enforces the observation layer’s quality gate.

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

[0.654, 0.703] on $n=1402$ completed calls; DeepSeek 0.540 [0.518, 0.561]; Gemini 0.359 [0.338, 0.380]). Vendor ranking within Ungated shifts relative to the 100-day sample (DeepSeek is less cautious at scale), but no vendor matches Compiled. Where Raw is trusted, over-trading remains sharp: GPT and Gemini abstain at 0.000 [0.000, 0.002]; DeepSeek stays cautious at 0.748 [0.728, 0.767] ($n=1925$). Because the full OOS window is entirely thin, Compiled CE is identically zero by construction; thick-day economics are deferred to Sec. 5.4.

Blind arm. On the same 100 asset-days, the no-feed prompt “Today is d . How should I trade BTC?” yields abstention on 100% of days for all four models (CE = 0; zero directional actions). Without an observation layer, the models refuse. The costly failure mode is the middle case: the same model given a Raw fragment—half-baked context—starts trading (abstain as low as 0.04) and loses. Compiled matches Blind’s refusal on thin days by the quality gate, while still

Table 6: Full-OOS band-thick / open slice (3/3 ready $\equiv$ WMI $\geq$ 0.05). Production Compiled abstains 1.0; Ungated CE point estimates are positive (bootstrap CIs include 0); Raw over-acts.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	-0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	-0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	-0.25

*Raw GLM $n=398$; Ungated GLM $n=932$; others $n=1224$. Four-vendor mean CE $\approx +0.29$;

block-bootstrap mean $\approx +0.31$ ($B=5$, 999 reps; one-sided $p \approx 0.21$ vs. CE ≤ 0).

exposing the structured world bundle that RQ2 and RQ3 show is usable when support improves.

Band-thick / open-slice days. Archive readiness alone is not a refuse policy. On full-OOS *band-thick* days (3/3 ready; 61% of OOS, $n \approx 1224$; $\equiv$ WMI ≥ 0.05), production Compiled still abstains at 1.0 (WMI < 0.2). Ungated on the same days keeps compiled content without the hard boolean: mean abs. ≈ 0.48 , CE positive for every vendor (four-vendor mean $\approx +0.29$; Table 6). Block bootstrap ($B=5$, 999 reps) CIs include zero (bootstrap mean CE $\approx +0.31$, one-sided $p \approx 0.21$); we report a content lower bound, not a trading edge, and *not* Compiled-open (the frozen prompt still forces abstain when the world is judged thin even if the boolean were false). Raw remains risky: GPT and Gemini abstain at 0.0 with CE -0.28 and -0.25 .

Inside the Ungated arm. Ungated rationales cite disclosed world fields on 53–93% of days, so models *read* the observation layer’s bundle; they differ in acting on it. Vendor-specific caution—not missing disclosure—drives the gap the hard quality gate closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only mocks ignore thin-world flags (thin-abs. 0). Offline results are protocol checks; live vendors are the main claim.

When would the layer open? Compiled never trades while max WMI (≈ 0.093) stays below the production threshold 0.2: a scarce-support stress test, not a permanently closed layer. A WMI gate of 0.05 would clear the boolean on $n=1224$ open days (ACWMI already ≥ 0.25), but would *not* clear the Compiled prompt’s separate “world is thin” clause. Live evidence on that slice is Ungated content (Table 6) plus the no-LLM rule vs. momentum ($\Delta\text{CE}=0.315$, $p=0.034$; Table 9). Denser vintages can raise the WMI floor without changing layer semantics.

5.3 RQ2: Seeing the market—grounding workflow

If the observation layer works, public LLMs should *see* what the world bundle discloses—not only obey a refuse bit. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report $\text{sign}(\text{macro_tilt})$ and $\text{sign}(\text{alt_tilt})$, and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 7: with the Compiled bundle, mean ready-band F1 0.97, missing-band

Table 7: Grounding workflow (50 days): ready / missing / tilt signs.

Model	Ready F1		Missing F1		Tilt-sign acc.	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency judgments alone are weak once completeness is scoped to real archive bands—models may call a 3/3-ready day “sufficient” while WMI still marks the world thin. The sharper claim is verifiability: only the compiled world bundle lets the model state what is ready, missing, and signed, in a form that can be checked. This is the analyst-facing face of the observation layer and the counterpart of Prop. 3.5’s missingness term.

5.4 RQ3: Secondary content probe

RQ1–RQ2 ask whether the observation layer is legible and safely gated. RQ3 asks whether the compiled tilts inside the world bundle are economically nonempty. A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt —no LLM in the loop. Out-of-sample (199 days $\times$ 10 assets) this content rule beats the baselines (Table 8, Figure 4): Sharpe 0.764 / CE +0.130 versus momentum (0.096, -0.206) and buy-and-hold (-1.404 , -1.163). The pre-specified contrast mechanism – momentum is significant under block bootstrap ($\Delta\text{CE}=0.334$, $p=0.034$). ΔCE is positive for all ten assets (Figure 4, right). LOBO (Prop. 3.8) attributes the gain to band content (deleting macro/alternative collapses to momentum); gaps survive 10–25bps costs. A 2017–2026 audit with non-vintaged proxy tilts shows no edge ($\Delta\text{CE} \approx 0$, $p=0.81$): value appears where the vintage-compiled world exists. The result supports nonempty fields in the observation layer, not an LLM trading ranking.

Within-archive dense / open world. Among the three archive bands, 61% of OOS asset-days are band-thick (3/3 ready) and 39% are band-thin (exchange gap; Figure 2). On this panel, band-thick coincides with counterfactual open days (WMI ≥ 0.05). Table 9: on that open/thick slice the content rule acts and beats momentum ($\Delta\text{CE}=0.315$; CE +0.061 vs. -0.254). On exchange-gap days the rule abstains more often, so its CE is high while ΔCE vs. momentum shrinks to 0.129—complementary to RQ1’s refuse-under-thin test, not a claim that thin days are more profitable to trade.

5.5 Discussion and threats

What the results mean. Three facts carry the paper, in product order. (i) *The compiled world is legible*: grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. collapse under Raw (RQ2)—the observation layer makes the market visible to public LLMs. (ii) *Thin days must refuse*: soft “consider abstaining” text does not; Compiled’s hard quality gate does, across live vendors (RQ1). Disclosure without enforcement is unstable on thin mixes (Ungated ≈ 0.68 ;

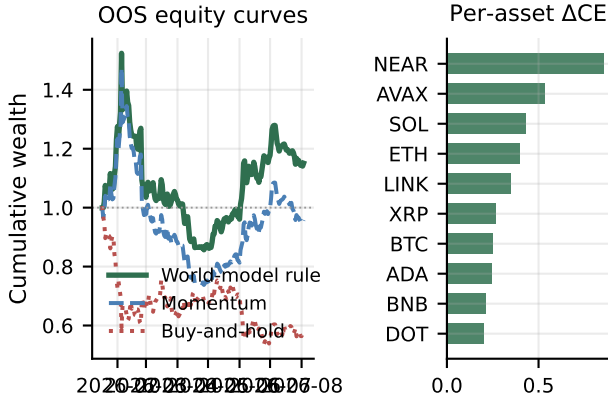
Table 8: Secondary content probe: world-model rule vs. traditional baselines (no LLM).

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	0.764	+0.130
Mechanism – Momentum	—	0.334 ($p=0.034$)

Table 9: Open/dense split of the content probe (OOS). Open = $WMI \geq 0.05 \equiv 3/3$ bands ready.

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick ($n=1224$)	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

*Block-bootstrap ΔCE (matches Table 8); raw mean gap $0.130 - (-0.206) = 0.336$.

**Figure 4: Secondary content probe (no LLM). Left: equity curves; right: ΔCE vs. momentum positive for every asset.**

GLM 0.86 vs. Gemini 0.43) and often destroys CE versus cash when the model acts—half-baked context harms decisions. (iii) *Open-slice fields are a nonempty lower bound*: Ungated CE is positive for all four vendors on $WMI \geq 0.05$ (Table 6; four-vendor mean $\approx +0.29$, bootstrap mean $\approx +0.31$, CIs include zero), and RQ3’s no-LLM rule is significant vs. momentum. Raw and Blind show unstructured or empty context cannot replace a typed world bundle. We claim a crypto observation layer—not LLM trading SOTA, not a generative simulator, not a full causal theory of crypto markets.

Relative to FinMem /FinAgent [23, 24] and ICAIF retrieval, debate, and judge lines [1, 3, 18, 19], we contribute the observation compiler and quality gate those stacks assume—not another memory, search, or debate router. Relative to return-prediction LLM studies [14], we score seeing and safe refusal before PnL.

Design rationale. Three choices are deliberate. (i) Ship a world bundle, not a feature dump: completeness, honesty, and the hard abstain flag make quality machine-readable rather than a free-text

apology. (ii) Previous-close PIT over wall-clock streaming: every live decision cites a recomputable decision_asof, avoiding same-day look-ahead. (iii) Four arms instead of one Sharpe number: Compiled / Ungated / Raw / Blind separates gated observation, soft judgment on the same bundle, thin integrations, and no feed.

Threats to validity. Compiled thin abstention is quality-gate obedience by design; Ungated removes the boolean on purpose. The archive is scarce (max $WMI \approx 0.093$), so production Compiled never opens at 0.2. Open-slice Ungated is a content lower bound, not Compiled-open trading: the prompt’s “world is thin” clause can still force abstain if the boolean were cleared, and CE bootstrap CIs include zero. Mixed-sample act-conditional Ungated CE is negative for three of four vendors; open-slice point estimates are positive—regime matters. Vendors are flash/mini at temperature 0; GLM Raw full-OOS is incomplete; denser bands are absent. Raw is common mom5 wiring for the four-arm ladder, not dominance over tool-using agents.

Scope. We do not claim LLM trading SOTA or generative dynamics. Natural extensions include richer bands and thick-regime live trading once WMI leaves the scarce floor—decision/execution stacks can sit on top of this observation layer. Camera-ready will restore author identity and release the anonymized harness deferred during review.

6 Conclusion

Public LLMs need a crypto market they can safely observe before they decide. We contribute an observation layer: an anonymized state-compiler runtime that turns asynchronous multi-band evidence into a consumable PIT world bundle, and forces abstention when quality is insufficient so half-baked context cannot harm decisions. Live, the compiled bundle is legible (grounding ≈ 0.97 vs. Raw ≤ 0.23); the hard quality gate abstains 1.0 on every thin day while Ungated soft-judges at ≈ 0.68 ; Raw over-trades and loses; Blind refuses with no feed. Open-slice Ungated and a no-LLM content probe give a nonempty lower bound for the compiled fields, not a Compiled-open trading claim. The contribution is the observation layer—seeing plus quality-gated refusal—not a trading leaderboard.

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